

(Revision – 2021)
Subject Code:

Reg. No.
Signature.

Third Semester Diploma Examination in Printing Technology

MATERIALS FOR PRINTING AND PACKAGING-SET-I

[Maximum Marks: 75]

[Time: Hours]

PART – A

I. Answer all questions in one word or one sentence.

- 1 Surfactants in ink are used to reduce
- 2 Recall the specialty of scratch and sniff ink.
- 3 List out any four optical properties of paper.
- 4 Specify the major raw materials of paper.
- 5 Expand the abbreviation BOPP.
- 6 Define the state Plasma.
- 7 Expand the abbreviation LDPE
- 8 Specify the list of any four adhesives used in printing and packaging.
- 9 Illustrate the Cross-sectional view of- Flexo plate.

(9 x 1 = 9 Marks)

PART B

II. Answer any *eight* questions. Each question carries 3 marks.

1. Recall evaporation drying of ink.
2. List out the features and advantages of water based ink.
3. Recall the importance of pulp refining process.
4. Define- Printability Properties of paper. Give two examples.
5. Spell the importance of using plastics in packaging.
6. Explain the steps involved in foiling on plastics.
7. Spell the importance of blankets in offset lithographic printing.
8. Tell the light sensitive requirements of image carrier materials used in lithographic printing.
9. Specify the use and features of plate treating solutions.
10. Illustrate an extruder unit that making extrusion blow molded bottle in a single stretch.

(8 x 3 = 24 Marks)

PART C

Answer all questions. Each question carries 7 marks.

III. Describe ink milling process of a 3 roll mill with necessary sketch.

OR

IV. Differentiate UV and solvent based inks.

V. Illustrate a digester unit and explain chemical pulping process.

OR

VI. Illustrate a rotary drum washer unit and explain brown stock pulp washing process in detail.

VII. Explain the working principle of corona treatment done on plastics.

OR

VIII. Describe the features and advantages of Plastic corrugated boards used in printing and packaging.

IX. Explain about the purpose of use of Leather in printing applications.

OR

X. Explain about the features and advantages of aluminium foil in printing and packaging purposes.

XI. Explain about any four common ink related printing problems and their remedies.

OR

XII. Summarize the features, advantages and applications of Fluoresce ink.

XIII. With the sketch, explain in detail about the board making process.

OR

XIV. Clarify the fact that-“*pulp refining influences mechanical properties of paper*“.

(6 x 7 = 42 Marks)

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MATERIALS FOR PRINTING AND PACKAGING-SET-II

[Maximum Marks: 75]

[Time: Hours]

PART – A

I. Answer all questions in one word or one sentence.

- 1 Tell about the purpose of doing milling in ink manufacturing.
- 2 Define length of ink.
- 3 Expand CTMP.
- 4 Gsm of paper ranges from.....to.....
- 5 Expand BOPP.
- 6 Chemical formula of polyethylene is
- 7 Image area of lithographic plate is (Oleophilic/Hydrophilic).
- 8 Show the cross-sectional view of lithographic blanket.
- 9 “Higher the film thickness lower will be the deep drying of uv coat.” Justify the reason.

(9 x 1 = 9 Marks)

PART B

II. Answer any *eight* questions. Each question carries 3 marks.

11. Differentiate pigments and dyestuffs.
12. Recall and label an ink manufacturing Flow chart.
13. Explain about the working of disc refiner unit and mark its efficiency.
14. Recall the paper recycling workflow.
15. Tell about the purpose of doing plasma treatments on plastic packages.
16. Relate contact angle and wettability with surface treatment of plastics.
17. How does digital imaging done in laser printer?
18. “The dampening solution does enhance the desensitization of litho-plates”. Explain.
19. Spell the reasons of printability of metal substrates.
20. Relate porosity and smoothness with printability.

(8 x 3 = 24 Marks)

PART C

Answer all questions. Each question carries 7 marks.

III. Explain the paste ink manufacturing steps in detail.

OR

IV. Describe the basic functions and features of oil and solvents used in ink manufacturing.

Give four examples for ink vehicles.

V. Explain about various Finishing and Converting operations doing in paper manufacturing process.

OR

VI. Differentiate Mechanical and Chemical pulping.

VII. Correlate moisture content of paper with its dimensional stability.

OR

VIII. Demonstrate the influence of pulp refining with paper stiffness.

IX. Compare PE and PP based on printing and packaging aspects.

OR

X. With sketch explain the laminating process doing on plastics for packaging purpose.

XI. With sketch explain about the imaging features of gravure cylinder.

OR

XII. Distinguish between thermal plates and visible light plates.

XIII. Explain in detail about different Rheological properties ink.

OR

XIV. Explain about the features and advantages of any seven types of security inks.

(6 x7 = 42 Marks)

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